

S.No. : 593

NBS 4103

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B. Tech. Examination, 2025-26

(Odd Semester)

ENGINEERING CHEMISTRY

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

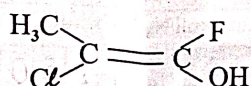
SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$

- (a) What is bond order of NO molecule?
- (b) Explain the basic principle of high performance liquid chromatography.
- (c) Give the name of polymer which is used for making helmet and bullet proof vests?

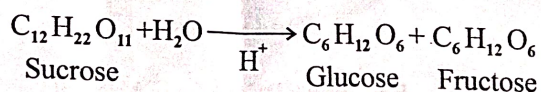
[P. T. O.]

- (d) Assign Z and E nomenclature :



- (e) Give any two example of biodegradable polymers.

- (f) In case of hydrolysis of sucrose :



What will the order of reactions?

- (g) Define chromophores.

- (h) Why is hardness expressed in terms of CaCO_3 equivalent?

SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) (i) Draw the molecular orbital diagram of O_2 and find out the bond order and its magnetic behaviour.

- (ii) KCl has rock salt structure and length of a side unit cell is 6.29082 cm, as determine by X-ray diffraction. Calculate density of unit cell.

- (b). Who gave the term green chemistry? Write twelve principles of green chemistry.

- (c) (i) Write short notes on the following :

(1) ISO / IEC 17025 : 2017

(2) Particulate matter

(3) Bathochromic shift

- (ii) Discuss non-stoichiometric defects in crystal.

- (d) What are persistent organic pollutants? Discuss in detail their major impacts on the environment and human health. Discuss about dirty dozens.

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Derive second order reaction when concentration of reactants are same.

- (b) (i) Explain structure of graphite. Explain the reasons for its electrical and lubricating properties.

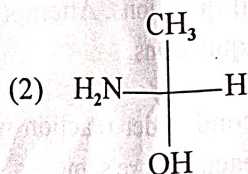
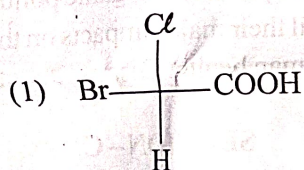
- (ii) Discuss thermotropic liquid crystals and its types.

[P. T. O.]

(e) (i) Discuss sol-gel method for the synthesis of nanoparticles.

(ii) The rate constant of a reaction is 1.5 sec^{-1} at 50°C and 4.5 sec^{-1} at 100°C . Calculate activation energy $R = 8.314 \text{ J/k/mol}$. $\log 3 = 0.4771$.

4. (a) (i) Assign R and S configuration to each of the following :



(ii) Write short notes on the following :

(1) Chirality

(2) Enantiomers

(3) Tautomerism

(b) What is optical isomerism. Give the stereoisomers of tartaric acid.

(e) Discuss various conformation of n-butane and explain their order of stability using energy profile diagram.

5. (a) Classify the polymer on the basis of :

(i) Mode of synthesis

(ii) Mode of stereochemistry

(b) Write the chemical structure and application of the following polymers :

(i) Nylon 6

(ii) Terelene

(iii) Bakelite

(iv) Teflon

(v) Nylon 6, 6

(c) Write about conducting polymers and its classification with its example.

6. (a) Describe the zeolite process for softening the hard water. What are the advantage and disadvantage of zeolite process.

[P.T.O.]

(b) Write short notes on the following :

(i) Types of transition in ultraviolet spectroscopy.

(ii) Biofuels and its importance.

(c) A sample of water on analysis was found to contain the following impurities in mg/lit.

Impurity	Ca (HCO ₃) ₂	Mg (HCO ₃) ₂	CaSO ₄	MgSO ₄
Quantity	10	8.5	12	14
Molecular weight	162	146	136	120

Calculate total hardness of water in mg/lit.
